DGPO inside πRL

Replacing πRL's Flow-SDE GRPO update with DGPO on $\pi_{0.5}$

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Goal

- Keep πRL's model, simulators, rollouts and reward; **replace only the policy update**: Flow-SDE GRPO $\rightarrow$ DGPO (arXiv:2510.08425).
- DGPO's promise: no stochastic policy needed, so rollouts can use the noise-free ODE sampler; reported $\sim$20–30$\times$ faster than Flow-GRPO on images.
- Stage 2: replace the binary success reward with Prism's success $+ \lambda \cdot$smoothness.

Experimental design

Model	$\pi_{0.5}$, starting from RLinf's released SFT checkpoint for each benchmark
Environments	LIBERO (Spatial, Long), MetaWorld MT50, ManiSkill pick-and-place, CALVIN ABC→D
Methods compared	**GRPO** (πRL, Flow-SDE noise) vs **DGPO** (ODE, or with SDE noise); optional Prism reward
Same for every method	SFT start, env count, groups of 8 rollouts per reset state, 12 h on 4×H100
Metric	held-out success with the **noise-free** sampler, 512 episodes per checkpoint ($\pm$2.2 pp)

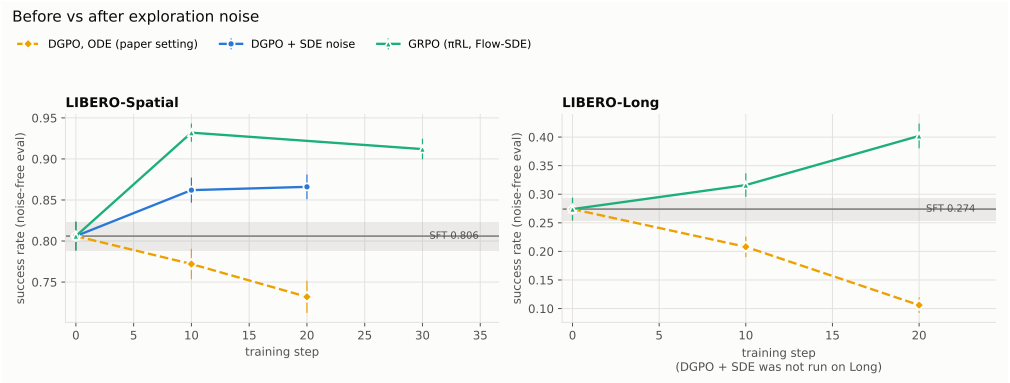
	LIBERO	MetaWorld	ManiSkill	CALVIN
Action	7-d end-effector delta + gripper	4-d xyz delta + gripper	7-d end-effector delta + gripper	7-d end-effector delta + gripper
SFT data	40 demos	2,500 demos	16,384 planner episodes	∼24 h play data
Reward	success 0/1	success 0/1	success + 0.1 grasp	subtask success
SFT success (ours)	0.806 Spatial, 0.274 Long	0.461	0.447	0.136

The one design choice that matters: exploration noise

	ODE (DGPO paper)	**SDE, every step**	**SDE, one step** (= GRPO's)
Noise during sampling	none	0.3 on each denoise step	0.5 on one random step
Rollout diversity	low	high	high
Used by	DGPO as published	DGPO "w/ SDE" variant	πRL's GRPO; DGPO control

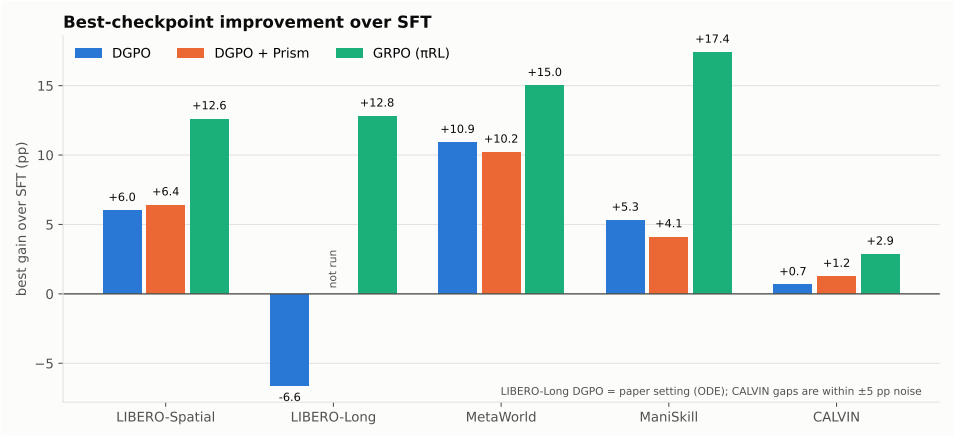
- GRPO needs the noise (it computes the likelihood of the noised step).
- DGPO does not need it in principle; we test all three.

Result 1: without noise DGPO gets worse, with noise it learns



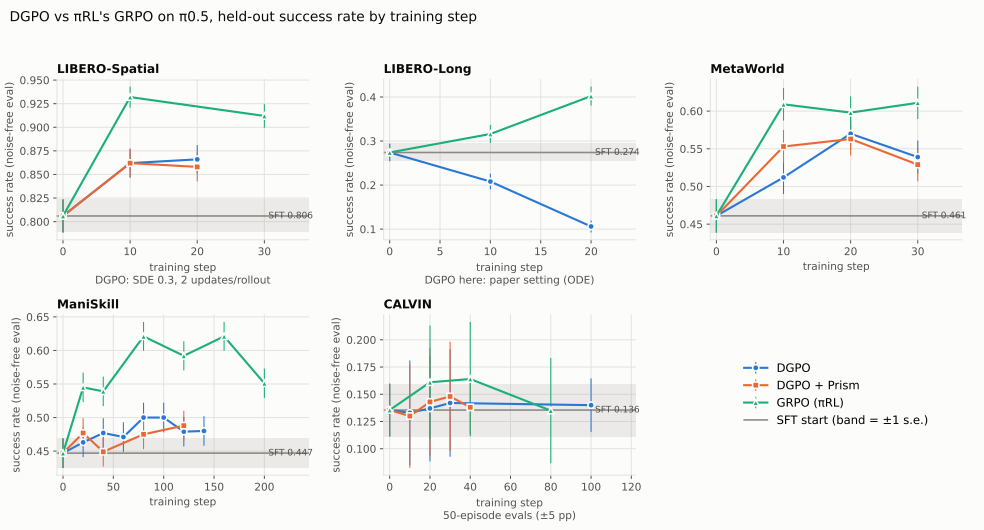
Grey band: SFT ± 1 s.e. The paper's ODE setting falls below SFT on both LIBERO suites.

Result 2: GRPO beats DGPO on every benchmark

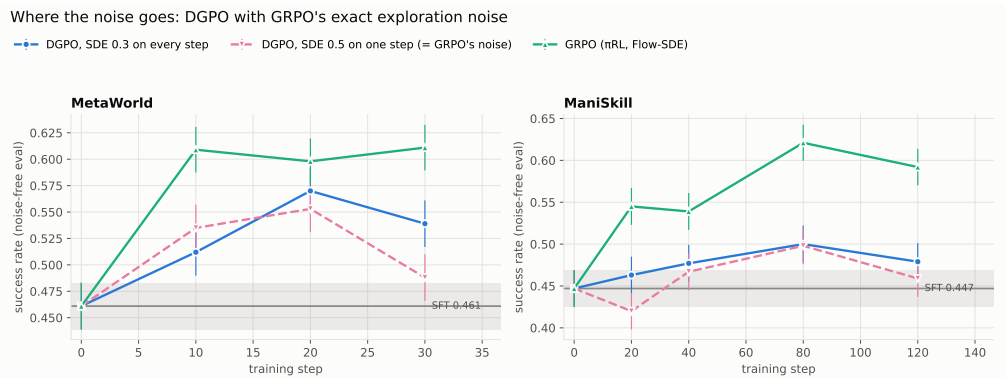


Best checkpoint vs SFT. DGPO = SDE noise on every step (LIBERO-Long: ODE only).

Result 3: learning curves



Result 4: GRPO's exact noise does not close the gap



Same rollouts as GRPO, different update only. A one-knob sweep (lr, β, updates/rollout, $t_{\min}$) on MetaWorld did not change this either.

Conclusions

- **DGPO as published (ODE) makes the $\pi_{0.5}$ policy worse** in πRL.
- With exploration noise DGPO beats SFT (LIBERO $+6$, MetaWorld $+11$, ManiSkill $+5$ pp), but **GRPO beats DGPO everywhere** ($+13$ / $+15$ / $+17$ pp).
- The gap is the update rule: our gradient matches the official DGPO code, tuning does not close it, and neither do identical rollouts.
- No speed-up: πRL's rollouts are simulator-bound, so DGPO's savings do not apply; GRPO's steps were 15–35% faster.
- The Prism reward adds nothing measurable; CALVIN shows no gain for any method.